Shear Pushout Universal Property

Date: 2026-06-21

Summary

Pass 86 states the correct universal property of the finite-adele shear extension

$$C_{\mathbb{Q}} = [\mathbb{Q} \to \mathbb{A}_f].$$

It is the pushout of

$$C_{\mathbb{Z}} = [\mathbb{Z} \to \widehat{\mathbb{Z}}]$$

along $\mathbb{Z} \hookrightarrow \mathbb{Q}$ and is initial among shear-marked quotient models with uniquely divisible, i.e. $\mathbb{Q}$-linear, kernel.

Main Claims

- **Initial pushout.** If $0 \to D \to E \to \epsilon \to 0$ is a shear-marked model receiving $C_{\mathbb{Z}}$ and D is uniquely divisible, then the map $\mathbb{Z} \to D$ extends uniquely to $\mathbb{Q} \to D$. Hence the model factors uniquely through $C_{\mathbb{Q}}$.
- **Finite-shadow stability.** The pushout does not create ordinary finite cokernels: integer residues already cover every checked finite quotient, so finite/Hausdorff shadows remain acyclic.
- **Torsion caveat.** The naive statement for arbitrary divisible kernels is false. The maps

$$\mathbb{Q} \to \mathbb{Q}/\mathbb{Z}, \qquad q \mapsto kq \bmod \mathbb{Z}$$

  restrict identically on $\mathbb{Z}$ but differ on fractions. Thus torsion divisible summands must be excluded or given extra shear data.

Machine Verification

`code/scripts/check-pass86.py` produced `artifacts/reports/pass86-shear-pushout-universal-property-check.json` with overall PASS. It checks bounded denominator localizations, finite residue acyclicity, unique factorization through checked $\mathbb{Q}$-vector targets, and the $\mathbb{Q}/\mathbb{Z}$ counterexample to naive divisible-kernel initiality.

Next Use

Pass 87 should upgrade this finite certificate to a mapping-space statement in $D(\text{Solid})$, identifying the homotopy fiber of shear-marked maps out of $C_{\mathbb{Q}}$ and deciding how torsion-divisible summands are excluded or decorated.